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# -----
banner("STEP 5 (bonus) - Causal models (GPT-2) only see the LEFT context")
# -----
# In both sentences the words BEFORE "bank" are identical; the disambiguating words
# come AFTER. A bidirectional model can use them, a causal (decoder-only) model cannot.
s1 = "The bank was flooded by the river."
s2 = "The bank was robbed by two men."
print(f"  A: {s1}\n    B: {s2}\n")
```

```
for name in ["bert-base-uncased", "gpt2"]:
    tok = AutoTokenizer.from_pretrained(name)
    mdl = AutoModel.from_pretrained(name, output_hidden_states=True).eval()
    va, ta = target_vectors(s1, TARGET, tok, mdl)
    vb, _ = target_vectors(s2, TARGET, tok, mdl)
    kind = "bidirectional" if "bert" in name else "causal (left-to-right)"
    print(f"  {name:18s} [{kind:22s}] cosine('{TARGET}' in A, in B) at last layer = "
          f"{cos(va[-1], vb[-1]):.4f}")
```

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=====
STEP 5 (bonus) - Causal models (GPT-2) only see the LEFT context
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```
  A: The bank was flooded by the river.
  B: The bank was robbed by two men.
```

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Loading weights: 100%|████████████████████| 199/199 [00:00<00:00, 14294.93it/s]
  bert-base-uncased [bidirectional] cosine('bank' in A, in B) at last layer = 0.6016
Loading weights: 100%|████████████████████| 148/148 [00:00<00:00, 7796.01it/s]
  gpt2 [causal (left-to-right)] cosine('bank' in A, in B) at last layer = 1.0000
```

GPT-2 gives exactly 1.0000: when it reads 'bank' it has not seen 'river' or 'robbed' yet.
The meaning is resolved later, in the vectors of the FOLLOWING tokens.